Coursework Research Project for GGS366 Spatial Computing

Submission Date: Monday, December 7th, 2026

To succeed in GGS366 it is important to submit a competent research project on a topic of your choosing. This coursework will consist of submitting a research paper which utilizes spatial computing on a topic of your choice.

The aim of this exercise is for each class participant to understand and apply the spatial computing concepts covered in a piece of analytical research, thereby achieving the course learning objectives.

To recap, the learning objectives for GGS366 are as follows:

1. Understanding the basics of computer programming (e.g., variables, functions, iteration etc.), as well as different design approaches (scripting versus object-oriented), with relevance to the spatial sciences.
2. Developing critical thinking skills with regard to spatial computing uses, design approaches, and methodological choices.
3. Mastering important programming applications (e.g., Google Colab) and key spatial computing packages (e.g., shapely, geopandas etc.).

Students are allowed to choose whichever programming method best suits their topic. Seek advice if you are unsure which path to take. Indeed, plenty of time is allocated in the course to help refine your project. Tasks set in class will complement the necessary processing steps for your chosen topic. Those who start early have a much higher probability of success.

Students are advised to consider this exercise as a piece of work which constitutes a potential job market paper, consequently demonstrating key competences when applying for future positions beyond GMU. Indeed, the most successful students can potentially convert this research into future funded projects.

The project requirements include:

- Submission of a scientific research paper which utilizes spatial computing techniques with a total paper length of at least 2,000 words (not including references) and an associated Python codebase capable of answering your research question(s).
- Students are encouraged to focus on their key current/future interests (e.g., relevant geospatial aspects of society, the economy and/or the environment).
- The paper should be submitted on MyMason Canvas (as a pdf document), and also uploaded to a GitHub repository with the developed code. LaTeX/MarkDown documents can also be submitted if you prefer, just make sure to provide a .pdf file of the final submission. Submissions not in .pdf, will not be graded (as they cannot be readily viewed in the Canvas web interface).

The paper needs to include:

- A properly written research abstract which summarizes the paper, including the motivation, research question, results, and findings (5 points).
- An introductory section which provides background information, the motivation for the analysis and a stated research question(s) which the analysis aims to answer (5 points).
- A comprehensive literature review on your chosen topic summarizing past theoretical and empirical research in this area, with at least 10 peer-reviewed citations (but more are better) (10 points).

- A high-quality methodology section which details the data sources and spatial computing steps involved in the analysis. This must include a box diagram illustrating the sequencing of the code-based processing steps, from input data to results output (10 points).
- Written results of the analysis, including panel plot maps, graphs (e.g., using Matplotlib) or other data chart types (10 points).
- A discussion section which critically evaluates the ramifications of the results in relation to the research question(s) specified in the introduction. Areas of future research could also be discussed. There must be a subsection on the limitations of the analysis (10 points).
- A conclusion section containing a summary of the purpose of the paper, and then the main findings (5 points).
- A fully documented bibliography which states the citations used in the paper (10 points).
- Submission of the Python spatial computing code on a GitHub repository (35 points).

Should a submission not meet any of the key criteria, then this will have a subsequent impact on the overall grade of the final project.

Importantly, please remember this is a spatial computing class and you will be graded on the Python code you write to undertake a piece of spatial processing. Writing a paper using a standard piece of GUI GIS software (e.g., ESRI ArcGIS /QGIS) will likely result in failing the class.

Finally, students may use AI tools in accordance with the course (and GMU) AI policy. Yet, students remain responsible for understanding, testing, validating, and documenting all submitted material. The paper must include a brief AI-use statement identifying the tools used and explaining how they contributed. Undisclosed AI-generated material, fabricated citations and other materials, or code that the student cannot explain may constitute an Honor Code violation and will be reported. **It is particularly important that the write-up of the report is written in your own words and not copied verbatim from an AI (so you develop your critical thinking skills). Your understanding will be cross-examined during the final presentation/oral defense. Additionally, AI writing is not enjoyable to read, so please do not submit this content here for evaluation.**